

scenario 1.py - C:/Users/ANJANI PRIYA/AppData/Local/Programs/Python/Python313/scenario 1.py (3.13.3)

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```
import numpy as np

house_data = np.array([
    [3,1400,7500000],
    [5,2200,12500000],
    [4,1800,9800000],
    [6,3000,18500000],
    [2,1100,5200000]
])

# Bedrooms > 4
houses = house_data[house_data[:,0] > 4]

avg_price = np.mean(houses[:,2])

print("Average Sale Price =", avg_price)
```

Ln: 17 Col: 0

IDLE Shell 3.13.3

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```
Python 3.13.3 (tags/v3.13.3:6280bb5, Apr 8 2025, 14:47:33) [MSC v.1943 64 bit (AMD64)]
on win32
Enter "help" below or click "Help" above for more information.

>>>
= RESTART: C:/Users/ANJANI PRIYA/AppData/Local/Programs/Python/Python313/scenario 1.py
Average Sale Price = 15500000.0
>>>
```

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scenario 2.py - C:/Users/ANJANI PRIYA/AppData/Local/Programs/Python/Python313/scenario 2.py (3.13.3)

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```
import pandas as pd
from sklearn.linear_model import LogisticRegression

# Training Data
data = {
    'Num_Links': [1, 6, 0, 4, 2],
    'Num_Caps_Words': [3, 15, 1, 10, 5],
    'Email_Length': [120, 300, 80, 250, 160],
    'Spam': [0, 1, 0, 1, 0]
}

df = pd.DataFrame(data)

# Features and Target
X = df[['Num_Links', 'Num_Caps_Words', 'Email_Length']]
y = df['Spam']

# Train Logistic Regression Model
model = LogisticRegression()
model.fit(X, y)

# New Email Details
links = int(input("Enter Number of Links: "))
caps = int(input("Enter Number of Capitalized Words: "))
length = int(input("Enter Email Length: "))

# Create DataFrame for Prediction
new_email = pd.DataFrame(
    [[links, caps, length]],
    columns=['Num_Links',
            'Num_Caps_Words',
            'Email_Length']
)

# Prediction
prediction = model.predict(new_email)

if prediction[0] == 1:
    print("Predicted Result : Spam Email")
else:
    print("Predicted Result : Not Spam Email")
```

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IDLE Shell 3.13.3

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```
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on win32
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>>>
= RESTART: C:/Users/ANJANI PRIYA/AppData/Local/Programs/Python/Python313/scenario 2.py
Enter Number of Links: 5
Enter Number of Capitalized Words: 8
Enter Email Length: 200
Predicted Result : Not Spam Email

>>>
```

Ln: 9 Col: 0

NIFTY +0.68%

Search

10-07-2026 09:32

SCENARIO 4.py - C:/Users/ANJANI PRIYA/AppData/Local/Programs/Python/Python313/SCENARIO 4.py (3.13.3)

File Edit Format Run Options Window Help

```
import pandas as pd
from sklearn.linear_model import LogisticRegression

# Dataset
data = {
    'Num_Links': [1, 6, 0, 4, 2],
    'Num_Caps_Words': [3, 15, 1, 10, 5],
    'Email_Length': [120, 300, 80, 250, 160],
    'Spam': [0, 1, 0, 1, 0]
}

df = pd.DataFrame(data)

# Input Features
X = df[['Num_Links', 'Num_Caps_Words', 'Email_Length']]

# Output
y = df['Spam']

# Create and Train Model
model = LogisticRegression()
model.fit(X, y)

# User Input for New Email
links = int(input("Enter Number of Links: "))
caps = int(input("Enter Number of Capitalized Words: "))
length = int(input("Enter Email Length: "))

# New Email Data
new_email = pd.DataFrame(
    [[links, caps, length]],
    columns=['Num_Links',
            'Num_Caps_Words',
            'Email_Length']
)

# Prediction
result = model.predict(new_email)

if result[0] == 1:
    print("The Email is Spam")
else:
    print("The Email is Not Spam")
```

Ln: 17 Col: 0

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```
Python 3.13.3 (tags/v3.13.3:6280bb5, Apr 8 2025, 14:47:33) [MSC v.1943 64 bit (AMD64)]
on win32
Enter "help" below or click "Help" above for more information.

>>>
= RESTART: C:/Users/ANJANI PRIYA/AppData/Local/Programs/Python/Python313/SCENARIO 4.py
Enter Number of Links: 4
Enter Number of Capitalized Words: 10
Enter Email Length: 250
The Email is Spam

>>>
```

Ln: 9 Col: 0

NIFTY +0.68%

Search

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SCENARIO3.py - C:/Users/ANJANI PRIYA/AppData/Local/Programs/Python/Python313/SCENARIO3.py (3.13.3)

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```
import pandas as pd
from sklearn.linear_model import LinearRegression

# Dataset
data = {
    'Age': [2, 5, 3, 7, 1],
    'Kms_Driven': [25000, 60000, 40000, 90000, 15000],
    'Engine_CC': [1200, 1500, 1300, 1800, 1000],
    'Resale_Price': [650000, 420000, 520000, 300000, 720000]
}

df = pd.DataFrame(data)

# Features and Target
X = df[['Age', 'Kms_Driven', 'Engine_CC']]
y = df['Resale_Price']

# Train Linear Regression Model
model = LinearRegression()
model.fit(X, y)

# User Input
age = int(input("Enter Car Age (Years): "))
kms = int(input("Enter Kilometers Driven: "))
engine = int(input("Enter Engine CC: "))

# New Car Data
new_car = pd.DataFrame(
    [[age, kms, engine]],
    columns=['Age', 'Kms_Driven', 'Engine_CC']
)

# Prediction
predicted_price = model.predict(new_car)

print("Estimated Resale Price = ₹", round(predicted_price[0], 2))
```

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```
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on win32
Enter "help" below or click "Help" above for more information.

>>>
= RESTART: C:/Users/ANJANI PRIYA/AppData/Local/Programs/Python/Python313/SCENARIO3.py
Enter Car Age (Years): 4
Enter Kilometers Driven: 50000
Enter Engine CC: 1400
Estimated Resale Price = ₹ 495263.16

>>>
```

Ln: 9 Col: 0

Gold +0.69%

Search

ENG IN

09:34 10-07-2026

```
SCENARIOS.py - C:/Users/ANJANI PRIYA/AppData/Local/Programs/Python/Python313/SCENARIOS.py (3.13.3)
File Edit Format Run Options Window Help

df = pd.DataFrame(data)

# Features and Target
X = df[['Income', 'Credit_Score', 'Existing_Loans']]
y = df[['Approved']]

# Create CART Model
model = DecisionTreeClassifier(criterion='gini',
                               random_state=0)

# Train Model
model.fit(X, y)

# User Input
income = int(input("Enter Income per Month: "))
credit = int(input("Enter Credit Score: "))
loans = int(input("Enter Existing Loans: "))

# New Applicant Data
new_applicant = pd.DataFrame(
    [[income, credit, loans]],
    columns=['Income',
             'Credit_Score',
             'Existing_Loans']
)

# Prediction
prediction = model.predict(new_applicant)

if prediction[0] == 1:
    print("Loan Approved")
else:
    print("Loan Rejected")

# Display Decision Rules
print("\nDecision Rules:\n")
rules = export_text(
    model,
    feature_names=['Income',
                   'Credit_Score',
                   'Existing_Loans']
)

print(rules)

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on win32
Enter "help" below or click "Help" above for more information.

>>>
= RESTART: C:/Users/ANJANI PRIYA/AppData/Local/Programs/Python/Python313/SCENARIOS.py
Enter Income per Month: 50000
Enter Credit Score: 700
Enter Existing Loans: 1
Loan Approved

Decision Rules:

|--- Credit_Score <= 680.00
| |--- class: 0
|--- Credit_Score > 680.00
| |--- class: 1

>>>

Ln: 17 Col: 0
```